

Linking & Loading

- Famous helloworld.c example

```
#include <stdio.h>
int main()
{
    printf("helloworld :-)\n");
    printf("hello OS\n");

    return 0;
}
```

```
lih@lih:~/SF/OS_Demo/LinkLoad$ gcc -c helloworld.c ; ls
helloworld.c  helloworld.o
lih@lih:~/SF/OS_Demo/LinkLoad$ gcc -o helloworld helloworld.o ; ls
helloworld  helloworld.c  helloworld.o
lih@lih:~/SF/OS_Demo/LinkLoad$ ./helloworld
helloworld :-)
hello OS
```

Linking & Loading

- Famous helloworld.c example

```
#include <stdio.h>
int main()
{
    printf("helloworld :-)\n");
    printf("hello OS\n");

    return 0;
}
```

```
lih@lih:~/SF/OS_Demo/LinkLoad$ ls
helloworld.c
lih@lih:~/SF/OS_Demo/LinkLoad$ gcc -o helloworld helloworld.c ; ls
helloworld  helloworld.c
lih@lih:~/SF/OS_Demo/LinkLoad$ ./helloworld
helloworld :-)
hello OS
```

Linking & Loading

- Famous helloworld.c example using header

```
#include "Print.h"
int main()
{
    int a = 314;
    Print(a);

    return 0;
}
```

```
lih@lih:~/SF/OS_Demo/LinkLoad$ cat Print.h
#include <stdio.h>
int Print(int a);
lih@lih:~/SF/OS_Demo/LinkLoad$ cat Print.c
#include "Print.h"
int Print(int a)
{
    printf("helloworld => %d :-)\n", a);
}
```

Linking & Loading

- Famous helloworld.c example using header

```
#include "Print.h"
int main()
{
    int a = 314;
    Print(a);

    return 0;
}
```

```
lih@lih:~/SF/OS_Demo/LinkLoad$ cat Print.h
#include <stdio.h>
int Print(int a);
lih@lih:~/SF/OS_Demo/LinkLoad$ cat Print.c
#include "Print.h"
int Print(int a)
{
    printf("helloworld => %d :-)\n", a);
}
```

```
lih@lih:~/SF/OS_Demo/LinkLoad$ ls
helloworld2.c helloworld.c Print.c Print.h
lih@lih:~/SF/OS_Demo/LinkLoad$ gcc -o helloworld2 helloworld2.c
/tmp/ccnT6uWw.o: In function `main':
helloworld2.c:(.text+0x15): undefined reference to `Print'
collect2: error: ld returned 1 exit status
lih@lih:~/SF/OS_Demo/LinkLoad$ gcc -c helloworld2.c Print.c ; ls
helloworld2.c helloworld2.o helloworld.c Print.c Print.h Print.o
lih@lih:~/SF/OS_Demo/LinkLoad$ gcc -o helloworld2 helloworld2.o Print.o ; ls
helloworld2 helloworld2.c helloworld2.o helloworld.c Print.c Print.h Print.o
lih@lih:~/SF/OS_Demo/LinkLoad$ ./helloworld2
helloworld => 314 :-)
```

Return Status

- Famous helloworld.c example revisited

```
#include <stdio.h>
int main()
{
    printf("helloworld :-)\n");
    printf("hello OS\n");
    return 0;
}
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus$ gcc -o helloworld helloworld.c
lih@lih:~/SF/OS_Demo/ReturnStatus$ ./helloworld ; echo $?
helloworld :-)
hello OS
0
```